Project 05

i-SPINE

B-end / Surgical Assistance System / User Experience

BRIEF

This project originated from a collaboration with the **Spine Surgery Department** at Xiangya Hospital. Through research, we refined the procedure and methods for **spinal pedicle screw surgery** and designed an **assisted positioning system** along with its **corresponding devices.**

TEAM PROJECT (2 People)

The project was completed through close collaboration with a clear and detailed division of responsibilities.

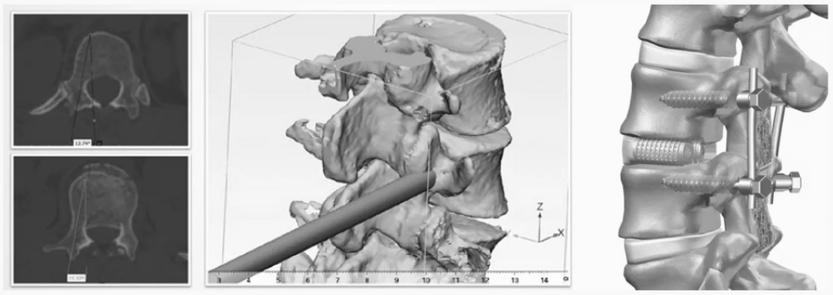
DURATION

Mar. 2025 – Jul. 2025

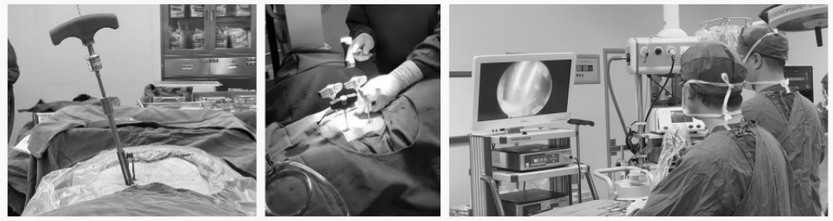


BACKGROUND

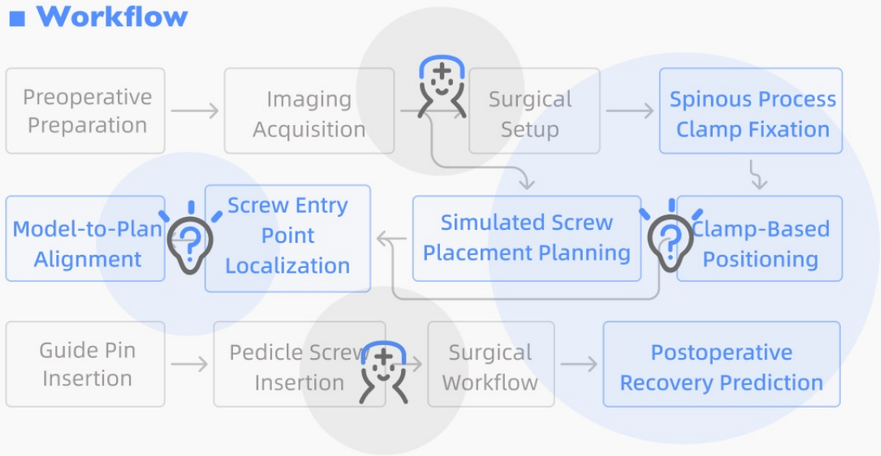
Spinal surgery requires **extremely high accuracy in screw placement**, as even minor deviations may cause **nerve or vascular injury.**



Traditional procedures rely heavily on **surgeons' experience and repeated X-ray verification**, resulting in **low efficiency, high radiation exposure, and increased error risk.**



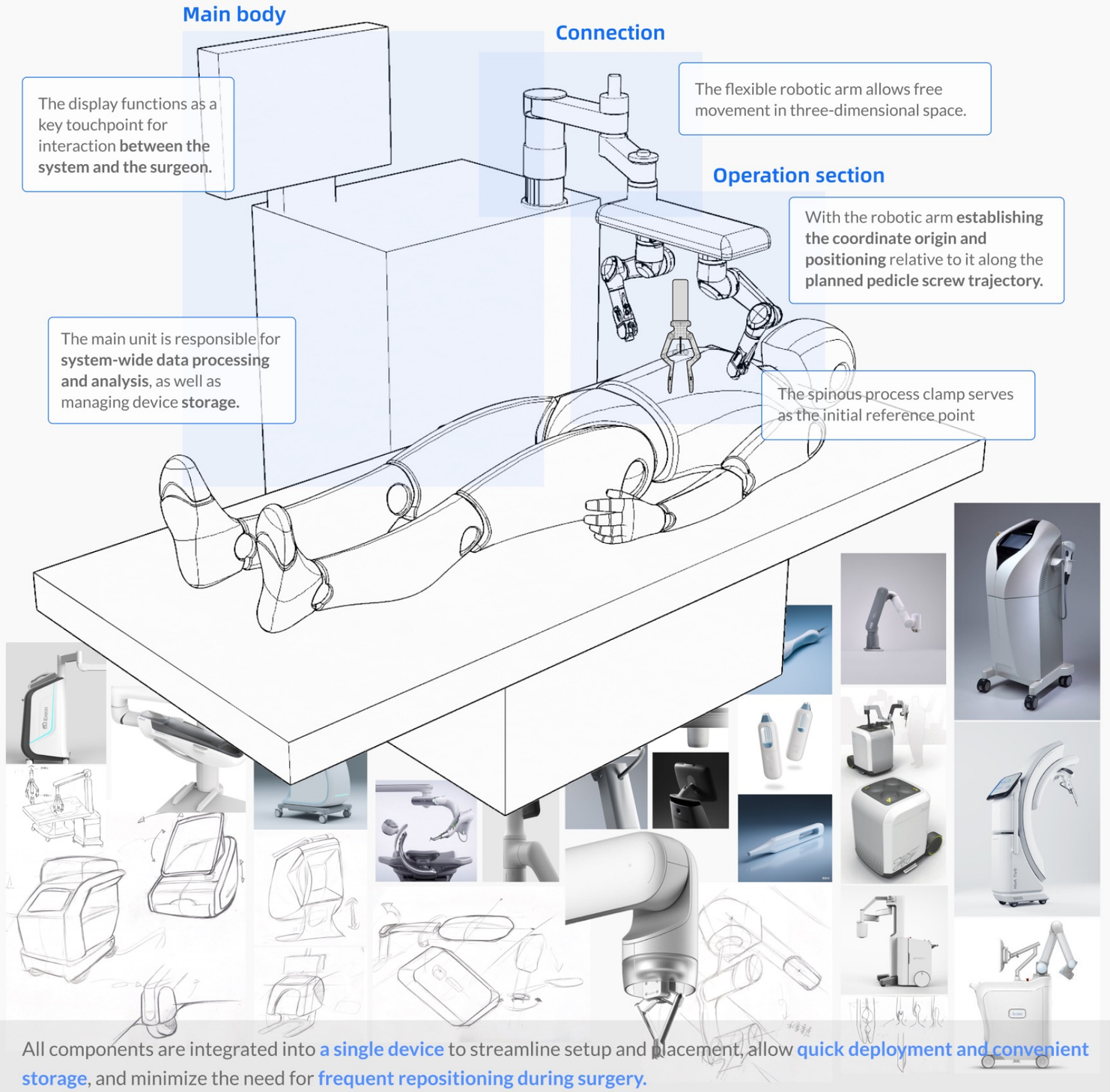
■ Workflow



■ Observation



IDEATION



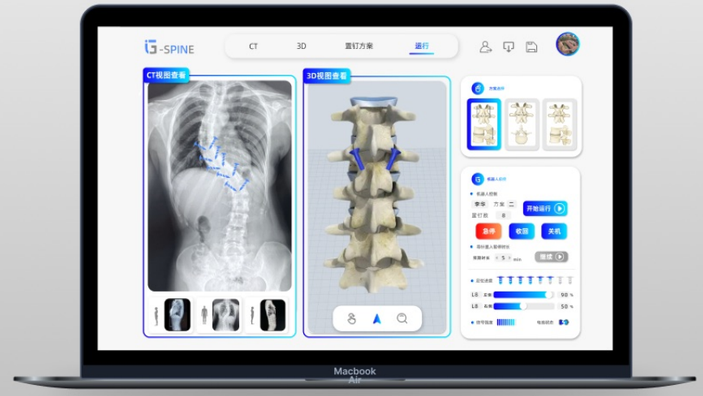
All components are integrated into **a single device** to streamline setup and placement, allow **quick deployment and convenient storage**, and minimize the need for **frequent repositioning during surgery.**

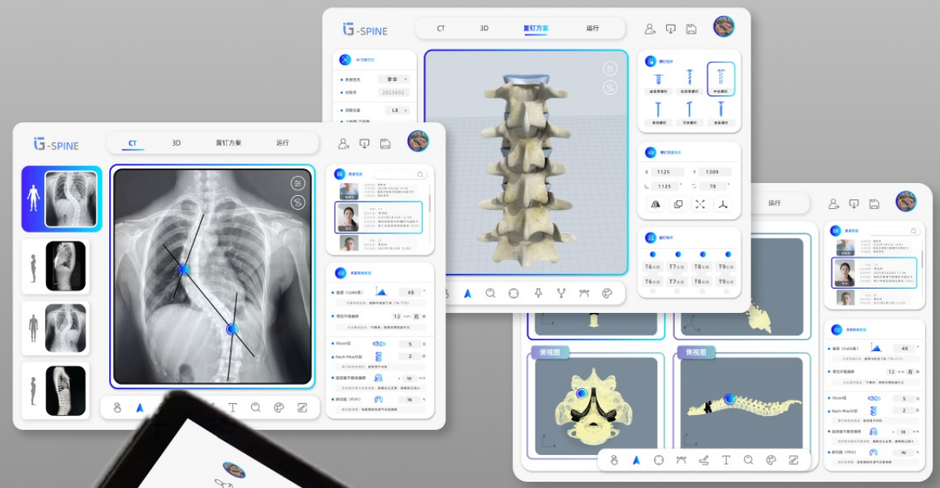
Through **a dual–mechanical-arm structure separating the connecting and operational modules** to achieves **full-angle, precise, and stable positioning** without optical tracking, reducing surgical invasiveness and cost.

OUTCOME

■ QR Code of the youtube video









■ Function

A three-dimensional model is generated through scanning to **define the specific pedicle screw placement plan and sequence**, and to establish the corresponding spatial coordinate system.

Through coordinate system computation, G-Spine accurately aligns with the predefined screw insertion points and directly provides the **surgeon with a guided channel** for positioning needle insertion.

The user interface provides intuitive and accessible touchpoints for **operating both the device and the overall system.**

USAGE









■ Technology

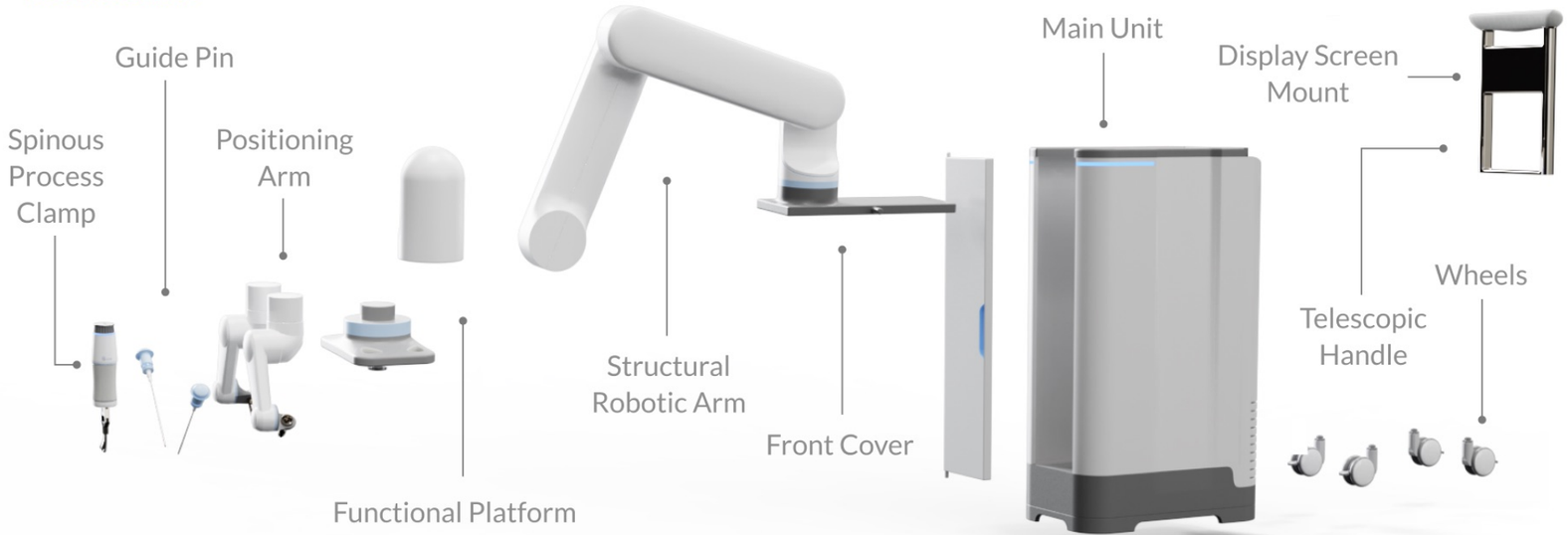
Spinous process clamp positioning origin:
Establish a spatial coordinate system through machine learning and define the coordinate origin.





Motor
Controls the opening angle and clamping force of the spinous process clamp

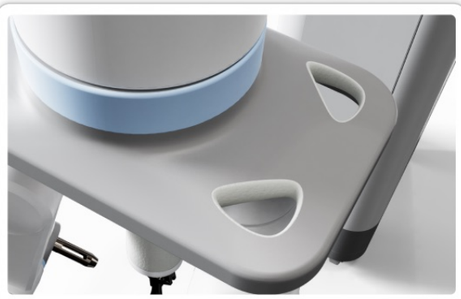
■ Structure



■ Details



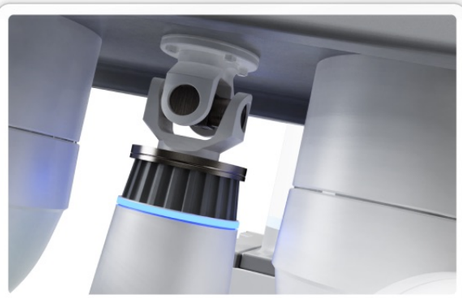
Precise clamping of the spinous process



Hollowed platform for comfortable grip



Expandable bracket for interface display

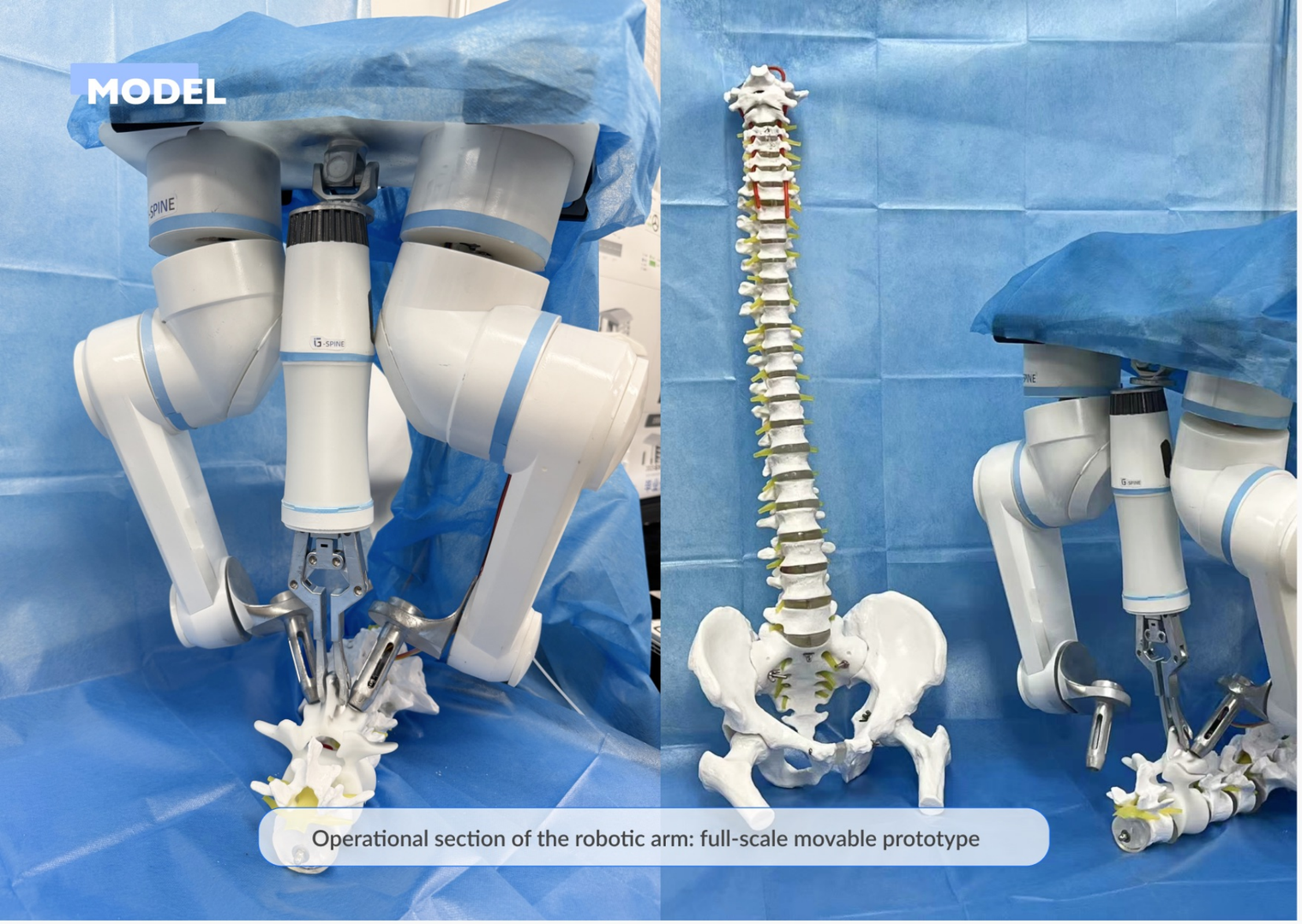


Universal fitting, multi-angle adhesion

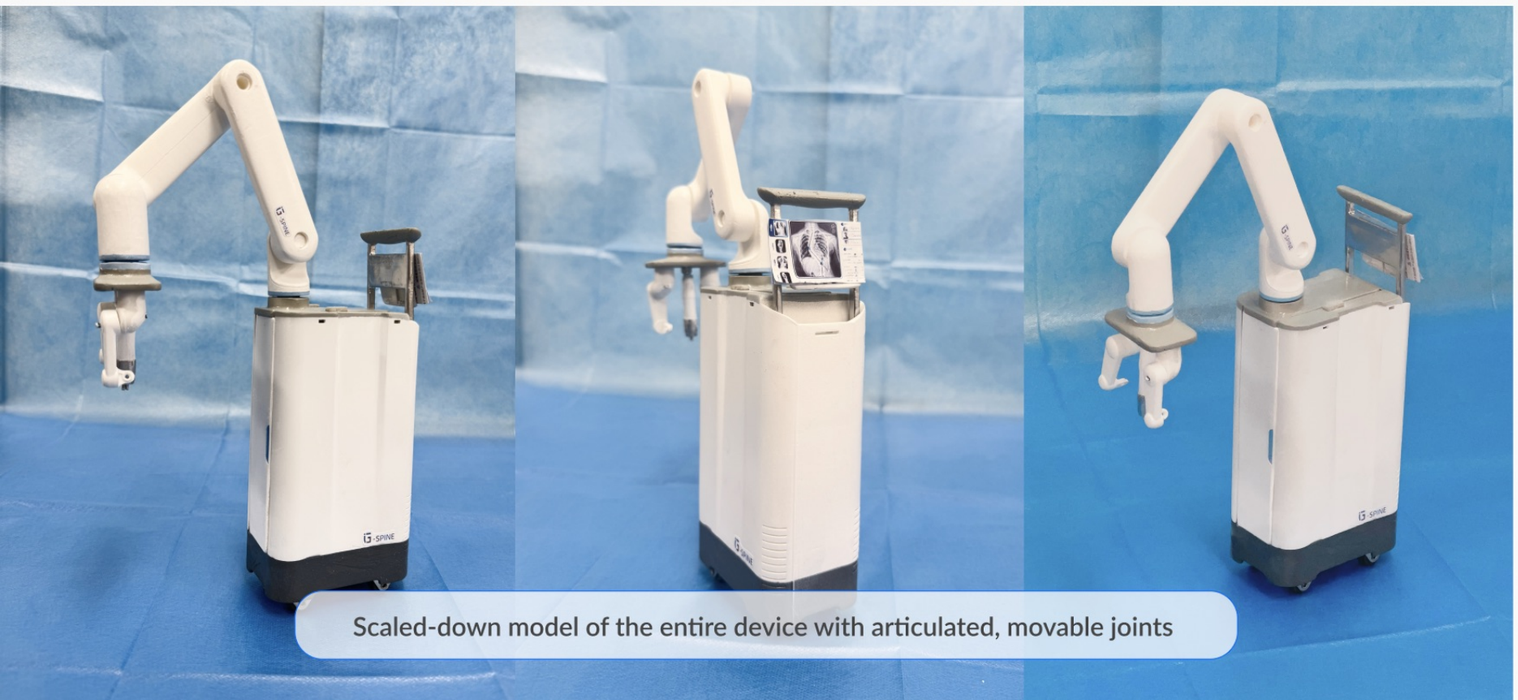


Universal limit wheel for stable placement

MODEL



Operational section of the robotic arm: full-scale movable prototype



Scaled-down model of the entire device with articulated, movable joints



The project exhibition area showcases the results of the course project, which have been recognized by the partners and mentors, and the project is continuously progressing.

REFLECTION

This solution is feasible, but it lacks a design for the entire system...

In the subsequent development, focus will be needed on **iterating the overall usage process and standard operating procedures**, as well as refining the model **dimensions for more detailed planning.**